

Assessment - Tool 1 - Analytical
Problem solving

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① Convert the following Infix expression into postfix notation

Given, $((A+B) \times C - (D-E) \times F + G)) / H$

Symbol	Stack	Postfix
(	(	
(	((	
A	((	A
+	((	AB
B	((	AB+
)	(	AB+
*	(	AB+C
C	(	AB+C
-	(	AB+C
(	(-	AB+C
D	(-	AB+C
-	(-	AB+C
E	(-	AB+C
)	(	AB+C
*	(	AB+C
F	(	AB+C
+	(	AB+C
G	(	AB+C
)	(	AB+C
)	(	AB+C
/	/	AB+C
H	/	AB+C

② Given Expression $((8+4) \rightarrow (1-3)/(2+1)) + (5 \times (6-2))$

Convert Infix to Postfix

Symbol	Stack	Postfix
(	(	
(	((	
8	((	8
+	((+	8
4	((+	84
)	(	84+
*	(*	84+
(	(*(	84+7
7	(*(	84+7
-	(*(-	84+73
3	(*(-	84+73
)	(*	84+73-
/	(/	84+73-
(	(/	84+73-*
2	(/	84+73-*2
+	(/+	84+73-*21
)	(/	84+73-*21+
)	(	84+73-*21+
+	+	84+73-*21+
(	+(	84+73-*21+/5
5	+(5	84+73-*21+/5
x	+(x	84+73-*21+/56
(	+(x(	84+73-*21+/56
6	+(x(6	84+73-21-/562
-	+(x(-	84+73-21+/562
2	+(x(-2	84+73-21+/562-
)	+(x(-	
)	+	

Evaluate Postfix using Stack

Symbol	Operation	Stack
8		8
4		8, 4
+	$8 + 4 = 12$	12
7		12, 7
3		12, 7, 3
-	$7 - 3 = 4$	12, 4
x	$12 \times 4 = 48$	48
2		48, 2
1		48, 2, 1
+	$2 + 1 = 3$	48, 3
/	$48 \div 3 = 16$	16
5		16, 5
6		16, 5, 6
2		16, 5, 6, 2
-	$6 - 2 = 4$	16, 5, 4
x	$5 \times 4 = 20$	16, 20
+	$16 + 20 = 36$	36

Final Result = 36